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DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

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TABLE OF CONTENTS

TITLE PAGE	1
GROUP LIST	4
ABSTRACT	5
CHAPTER ONE	6
INTRODUCTION	6
1.1 BACKGROUND TO THE STUDY	6
1.2 OBJECTIVES OF THE STUDY	8
1.3 RESEARCH QUESTIONS	9
1.4 SIGNIFICANCE OF THE STUDY	9
1.5 RESEARCH HYPOTHESIS	9
CHAPTER TWO	9
LITERATURE REVIEW	9
2.1 INTRODUCTION	9
2.2 SOCIAL MEDIA AND IT'S USAGE	9
2.3 RELATIONSHIP BETWEEN SOCIAL MEDIA USAGE AND IT'S IMPACTS ON STUDENTS ACADEMIC PERFORMANCE	10
2.4 SUMMARY OF LITERATURE REVIEW	11
CHAPTER THREE	12
METHODOLOGY	12
3.1 INTRODUCTION	12
3.2 THE STUDY AREA	12
3.3 POLPULATION AND SAMPLING TECHNIQUES.....	12
3.4 DATA COLLECTION PROCEDURE	12
3.5 RESEARCH INSTRUMENT	12
3.6 RESEARCH DESIGN	12
CHAPTER FOUR	13
ANALYSIS, DISCUSSION & PRESENTATION OF RESULTS.....	14

4.1 INTRIDUCTION	13
4.2 OPENNESS OF SOCIAL MEDIA USAGE	13
4.3 RELATIONSHIP BETWEEN SOCIAL MEDIA USAGE AND ACADEMIC PERFORMANCE	15
4.4 CONSTITUENTS OF SOCIAL MEDIA THAT IMPACT ACADEMIC PERFORMANCE	17
4.5 DISCUSSION OF RESULTS	19
CHAPTER FIVE	21
SUMMARY OF FINDINGS, CONCLUSIONS & RECOMMENDATIONS	21
5.1 SUMMARY OF FINDINGS	21
5.2 CONCLUSION	22
5.3 RECOMMENDATIONS	23
REFERENCES	24

THE IMPACT OF SOCIAL MEDIA USAGE ON ACADEMIC PERFORMANCE: A CASE STUDY OF KNUST STUDENTS

ABSTRACT

**This paper aims to assess the impact of social media usage on academic performance of
students of Kwame Nkrumah**

University of Science and Technology, Ghana. In this paper, the researchers are going to find out the real impact of social media usage on KNUST students' academic performance. Descriptive online survey questionnaire was conducted to show the real image of circumstances. Information on 333 students from all colleges of the university were randomly selected as the respondents of the study. It was conducted during the second semester in 2024. This study found out that social media is a part of students' life. Nonetheless, it is of little impact on students' academic performance. It helps to improve students' academic performance when it is used for academic purposes but tend to decrease students' grades when it is used for entertainment and leisure. In addition, through social media platforms students communicate with each other, seeking for career opportunities and social support and campus life and events. The study found that on average students spend 4.77 hours on social media with a standard deviation of 2.85 hours, indicating that there is some variability in how much time students spend on social media. Some students spend much less or much more time than the average.

Finally, it can be concluded that social media has a slight negative impact on students' academic performance. Usage of social media is important for students if they channel the time they spend on social media mainly on academic purposes.

Keywords: Technology, internet and social media platforms, KNUST students.

INTRODUCTION

The continuous and recent advancement in technology and digitalization in the 21st century has increased the usage of the internet. The internet provides an interactive form of media that allows users to interact with each other via various social media platforms.

*“The 21st century saw a huge increase in **social media** thanks to the widespread availability of the internet”.*

Some of the mostly used social media platforms include Facebook, YouTube, Instagram, X, WhatsApp, LinkedIn, TikTok, Snapchat, etc. When these social network sites were recognized, they came with two ways of accessing /performing them; online and offline. The online mode of usage is undoubtedly the

widely used and most predominant since most social media platforms requires an internet connectivity.

Social media has become an integral part of modern life, with billions of people around the world using various platforms to connect with others, share experiences, and access information 1

. While social media offers numerous benefits, concerns have been raised about its potential impact on academic

performance. Research has shown that excessive social media usage can lead to decreased attention span, increased stress

levels, and poor academic performance. However, social media also offers opportunities for learning, collaboration, and communication. This study aims to explore the complex relationship between social media usage and academic performance among KNUST students.

Social media is a modern technology which has different types of online platforms that helps people to connect one another from different parts of the world in a time. It helps people to save time and deal with others from a long distance. These online platforms have been used to build relationships between people from the time of its emergence (social media by Students (2017)).

Social media has helped us to communicate with one another on different contents, it is optional and on the base of our interest to select one. There are social media platforms to listen to music, to read books, to view photos and watch videos. Social media has made our lives much simpler and easier (Shensa, A & others (2020)).

Despite the high usage of social media among students, social media platforms have been linked to increased levels of anxiety and depression in students, which is effectively taking an impact on their academics. This study aims to investigate the relationship between social media usage and how it affects the academics of students in KNUST; with a focus on identifying potential risk factors and informing strategies on healthier social media engagement and how it will help improve the academics of students.

Objectives of the study

- To determine the factors that influence the usage of social media platforms among KNUST students
- To determine the social media platform that is mostly used by KNUST students
 - To check the relationship between social media usage and academic performance among KNUST students.
- To check how much time KNUST students spend on social media

Research questions

This paper aims to find out the influence of social media on KNUST students' academic performance. Specifically seeks to answer the following questions:

- What are the factors that influence the usage of social media platforms among KNUST students
- Which social media platform(s) is/are mostly used by KNUST students
- What is the relationship between social media usage and academic performance among KNUST students
- How much time KNUST students spend on social media

CHAPTER TWO LITERATURE REVIEW

The traditional form of education is shifting toward the use of e-learning in which much weight is given on the use of electronic

delivery of the content (**Allen and Seaman, 2008; Farid et al., 2018**). E-learning offers the online delivery of information, communication, education and training. By using e-learning approaches, learning and training can be carried out everywhere and anytime (**Tucker and Gentry, 2009**). E-learning education can be used effectively by adopting a systematic approach and ensuring that students are improving their

knowledge and skills (**Rothwell and Kazanas, 2011**). Classroom teaching allows the learners to focus on their core skills and

knowledge while online learning helps the students to revise their work and discuss it with their friends through online discussions. Social media platforms are playing a significant role in the elearning process and in the learning and teaching process (**Chang, 2016**)

Research has shown that social media sites have impacts on many aspects of our lives including education. Internet usage is beneficial in the context of education. It provides variety of

information sources globally and makes room to discuss and share experiences. The internet serves as a valuable resource to aid students in their research and projects. The internet has prepared possibilities for higher education.

The impact of social media on the different age groups has been widely discussed by various researchers in their studies. In addition, there exist so many studies in which the impact of social media on educational activities has been studied (**Abbas et al., 2019**). **Wickramanayake and Muhammad Jika (2018)**

reported the wide use and popularity of social media among undergraduate students for both education and entertainment purposes. They further reported that the use of social media enhances the learning skills in students. Social media is widely acceptable and useable among the higher education students.

The use of social media platforms by the masses serves as a key tool for communicating with friends, family, classmates, teachers, and so forth. The use of social media helps in collaborative learning. Students are motivated by social media to improve their learning and enhance their relationships (**AlRahmi, W., & Othman, 2013**). According to **Raymond Owusu Boateng & Afua Amankwaa, 2016**, social media has significant influence on students' academic performances. Communication settings has completely changed by social media. The study of **Fasae and Adegbilero-Iwari (2016)** reported Facebook as the most famous, while Google as the most beneficial social media

network among students in their educational activities. The use and impact of social media in education highlighted that by

using these platforms in educational activities, students get help to learn quickly and progress rapidly (**State of Victoria**

Department of Education and Early Childhood Development,

2010); An Australian Scenario. (**Orús et al. (2016)** reported that content creation in the form of videos positively impacts the “crosscurricular competencies” and leads to upgrading the academic performances of students. **Al-Sharqi et al. (2015)** were of the view that social media tools help to enhance

learning, help to generate novel ideas, but on the other side,

these tools can have a devastating effect on the academics of students as these tools can lead to distraction of student's

attention and can greatly minimize their interaction with friends and instructors. Hence, students can become potentially

addictive to these tools. **Jordaan and Surujlal (2013)** make the same claim that excessive use of social media sites can be a spoiler as students in their

teenage tend to become more addictive toward mobile phones rather than reading their books. In their study, **Karpinski and Duberstein (2009)** were also of the view that Facebook usage negatively affects the

academic performance of students of undergraduate and graduate levels. According to studies by several researchers, the use of social media helps its users especially students to be social, find new friends, get general information about daily life. Social media is necessary for students' engagements and achievements but it should be used with utmost responsibility.

The literature review on the impact of social media usage on the academic performance of students has shown significantly mixed arguments. However, there is a gap in the literature to be filled or otherwise be reported in the case of KNUST students about their usage of social media platforms and how it affects their academic performances. This research is purposely designed to help tackle that instance.

CHAPTER THREE

METHODOLOGY /MATERIALS AND METHODS 3.1 INTRODUCTION

The term methodology is a system of explicit rules and procedures in which research is based and against which claims of knowledge are evaluated (Ojo, 2009). Therefore, this section focuses on the research techniques adopted and used for this study with the aim of achieving the research objectives.

A survey method was employed where we designed a questionnaire for the data collection. The questionnaire was randomly issued to students of KNUST where the data was collected. Both qualitative and quantitative data was collected.

The questionnaire consisted of closed ended and Likert scale type questions. The study area is Kwame Nkrumah University of Science and Technology (KNUST), Kumasi. The population was all students of KNUST. The questionnaire was shared on various WhatsApp platforms of KNUST students. The questionnaire description and instructions specified that only KNUST students were eligible to participate. The participants were asked different questions about their demographics, purpose of using social media, time spent on social media and its significance, etc. A total of 333 responses were received for the analysis. The analytical tools employed in the data analysis are R (version 4.3.2), STATA and Statistical software for the social sciences (SPSS).

Some of the graphs were taken from the online questionnaire responses display.

CHAPTER FOUR

RESULTS AND DISCUSSIONS /DATA ANALYSIS INTRODUCTION

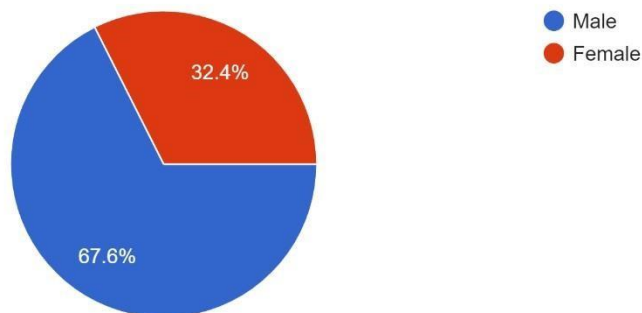
This chapter presents the analysis and discussion of data gathered from the study conducted at Kwame Nkrumah University of Science and Technology. Responses from all the respondents were examined, compiled and evaluated to answer the research questions asked at the beginning of the study. As far as possible, data were tabulated and displayed through tables with the aim of identifying and discerning any patterns that provided the best interpretation of the results of the study.

RESULTS

The results showed that 331(99.4%) of the respondents use social media while only two 2(0.6%) responded no, indicating that they do not use social media.

Figure 1.

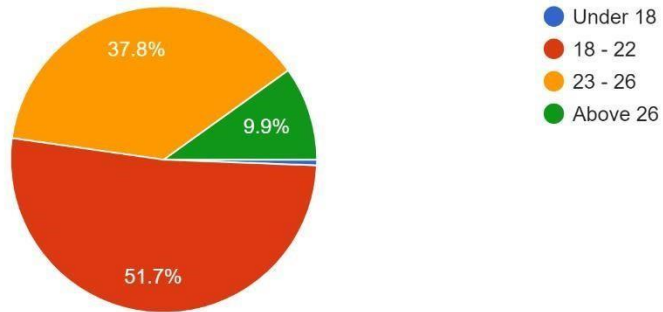
1. Gender
333 responses



This indicates that male students formed the highest part of our survey Figure 2.

2. Select your age category

333 responses



3. What is level of study

333 responses

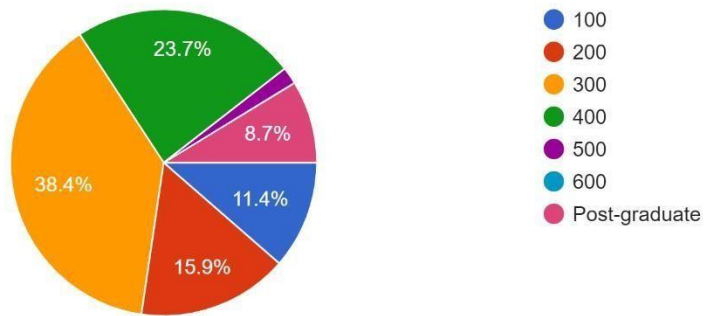


Figure 3.

4. Select your College

333 responses

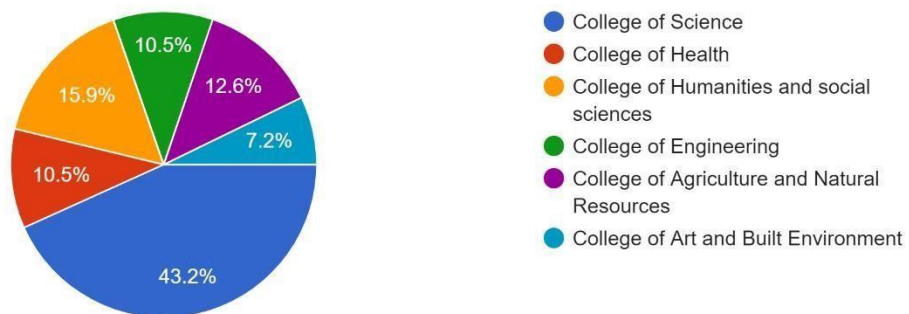


Table 4.0 Is there a relationship between the average time students spend on social media and their academic performance.

Method	p-value	Decision
Spearman	0.5273	Do not reject H_0
Pearson	0.1247	Do not reject H_0

From the results displayed on the above table, the Spearman correlation and Pearson correlation methods have provided evidence that there is a relationship between time spent on social media and academic performance among student. In order to check the direction of the relationship between time spent on social media and academic performance of students, a correlation analysis was performed on the two variables. This analysis came in two sides, the relationship between those who use social media platforms for academic performance and the relationship between those who use social media for entertainment and leisure.

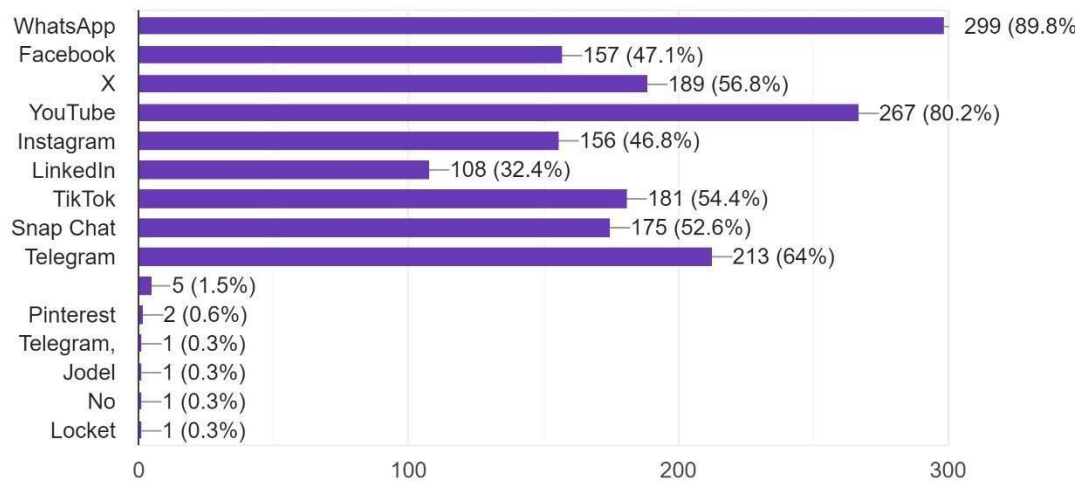
This overall analysis gave a correlation coefficient of -0.08443745 indicating a weak negative relationship between student academic performance and the time they spent on social media, this agrees with Paul A. Kirschner & Aryn C. Karpinski (2010, Aryn C. Karpinski & Adam Duberstein (2009).

This analysis also gave a correlation coefficient of 0.094645 indicating a weak positive relationship between student academic performance and the time they spent on social media for academic purposes. This means that there are other factors which are also responsible for student academic performance aside the time they spend on social media platforms. Further research can be conducted to explore more on the other factors.

Figure 4.

7. Which social media platform(s) do you use Select all that apply

333 responses



The high percentage of respondents about social media usage which has been found out by this study is a prove that most of the students use different types of social media in KNUST. Among them, WhatsApp and YouTube are the most used sites.

8. Which social media platform(s) do you use for academic purposes Select all that apply

333 responses

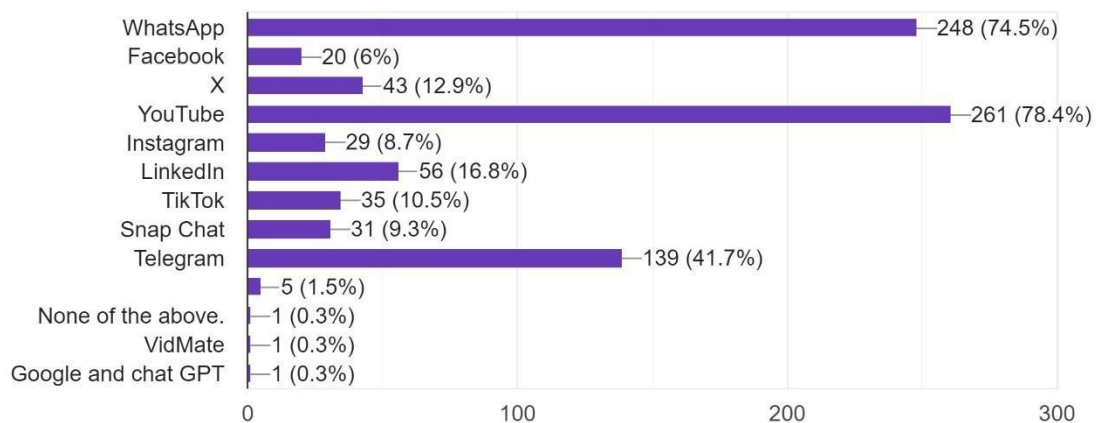


Figure 5.

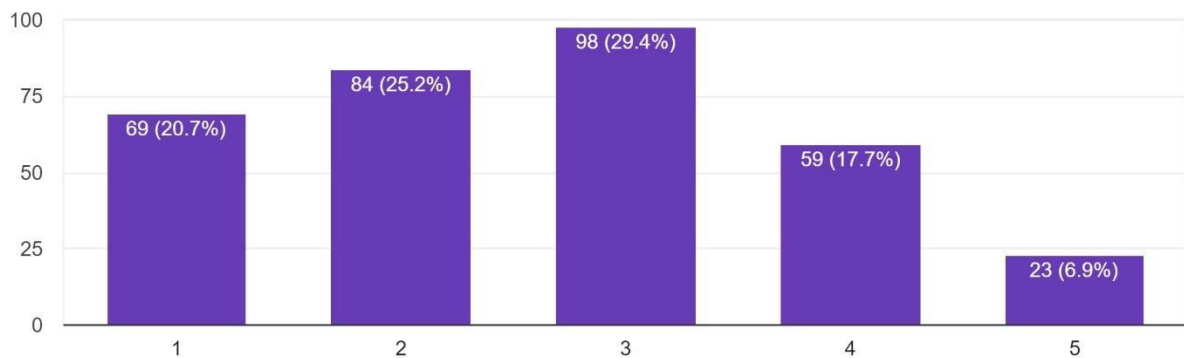
This above figure indicates that a huge number of the students who use social media for academic purposes heavily relies on YouTube 261(78.4%) followed by WhatsApp 248(74.5%). This can be attributed to the fact that YouTube provides some series of videos to explain concepts in various fields of academia.

ANALYSIS OF STUDENTS PERCEPTION ON THE TIME THEY SPEND ON SOCIAL MEDIA PLATFORMS AND THE IMPACT IT HAS ON THEIR STUDY TIME

Figure 6.

10. I use social media than I study

333 responses

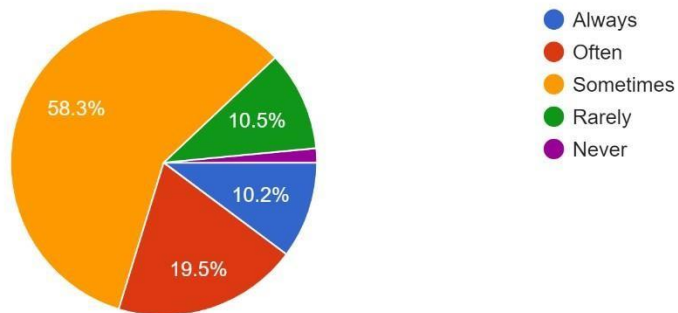


The Relative Perception Index (RPI) is approximately 2.65. This value is closer to 3 (Neutral), indicating that on average, respondents tend to be neutral or slightly agree with the statement "I use social media more than I study".

Figure 7.

14. How often do you feel distracted while studying as a result of social media notifications

333 responses



The above figure indicates that most students sometimes get distracted while studying as a result of social media notifications. This means that most students roam on social media platforms even during studies whilst a very small portion 10.5% never get distracted during their studies. This can further be attributed to the fact that most of the students 265(79.6%) indicated that they use social media to understand what they were taught in class. This means that, while they are trying to understand what they were taught in class via a social media platform they get some sort of notifications which are distractive.

Figure 8.

6. What drives your social media engagement

333 responses

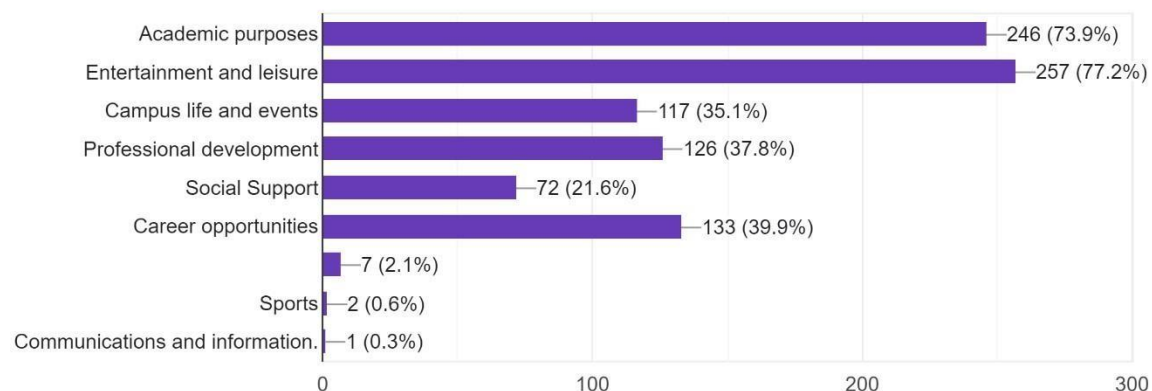


Figure 9 indicates that the number one reason why KNUST students use social media is Entertainment and Leisure followed by Academic Purposes while the other reasons recorded relatively small numbers as displayed by figure 9 above.

This further supports the reason for the weak negative correlation between student academic performance and time spent on social media.

Variable	value
mean	4.77
Standard deviation	2.85
Median	4.75
Trimmed mean	4.48
Median absolute deviation	2.59
Minimum	0.5
Maximum	16
Range	15.5
Skewness	1.21
Kurtosis	2.14
Standard error	0.16

How much time do students spend on social media

Figure 10.

The results suggest that a typical student spends about 4 to 5 hours on social media daily, with significant variation across the student population. The positive skewness indicates that while some students may limit their social media usage, others engage in it extensively, which could potentially affect their academic performance differently.

This data is critical for understanding the range of social media usage and identifying whether the amount of time spent could be correlated with varying academic outcomes, especially for those at the extremes of the distribution

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Social media has become an integral part of the people's life in today's world, where a variety of technologies are extensively accessible. Students is a part of society who actively and regularly use social media. This research was conducted to assess the impact of social media usage on KNUST students' academic performance. The findings of the study showed that usage of social media has a very weak negative impact on academic performance for students who use social media for entertainment and leisure. Social media help them to engage and understand what they were taught in class and also for entertainment and leisure. In addition, social media is an opportunity for students to understand academic materials related to the students' daily courses. However, the majority of students confirmed that they are neutral when they were asked the question "I use social media than I study" almost equal time on study and social media, but that students use social media for their academic purposes. It is worth indicating that, students get a little positive improvement in their grades when the time they spend on social media is for academic purposes and a little negative impact on their grades when they spend that time on social media for entertainment and leisure. This is evident from the results of our correlation analysis.

Therefore, it cannot affect their academic life negatively. The study also revealed that WhatsApp and YouTube are the most useful social media platforms among other types of social media which are used by students.

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APPENDIX

Below is a link to the online questionnaire that the reader can refer to know the type of questions that were asked during the data collection.

<https://forms.gle/WZyCCD7iqncp5Czq9>

https://docs.google.com/forms/d/e/1FAIpQLSc2mX8RZFJOEiSksnL7KPaa8RSNpuG5fl0eEZ17Zh_puwx8A/viewform?usp=sf_link

Embed HTML

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<iframe  
src="https://docs.google.com/forms/d/e/1FAIpQLSc2mX8RZFJOEiSksnL7KPaa8RSNpuG5fl0eEZ17Zh_puwx8A/viewform?embedded=true" width="700" height="520" frameborder="0"  
marginheight="0" marginwidth="0">Loading...</iframe>
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